Alayen Iraqi University

Engineering College/ AI Engineering Department

اشارات ونظم عملي ٥

ذكاء شعبة الثالثة

3rd year

Signals & Systems Lab

Lab (5)

Properties of Signals 1

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2.4 Properties of Signals

2.4.1 Periodic Signals

$x(t) = x(t+T), \quad \forall t \in \mathbb{R}.$

If T is the smallest positive value $\Rightarrow$ fundamental period of the signal

$x(t) = x(t+kT), \quad \forall t \in \mathbb{R}.$

For example
$x(t) = A\cos(\Omega t + \theta)$
$T = 2\pi/\omega$ fundamental period

$x[n] = x[n+N], \quad \forall n \in \mathbb{Z}.$ N positive integer number

$x[n] = \cos(\omega n + \varphi)$

A discrete-time signal $x[n]$ is periodic

if there are *integer* positive numbers m, N, such that

$$\omega = \frac{2\pi m}{N}$$

$x[n] = 2\cos(n/2 + \pi/4)$ is not periodic

$y[n] = 2\cos(n\pi/6 + \pi/4)$ is periodic $N = 12$ fundamental period

2.4 Properties of Signals

2.4.1 Periodic Signals

Example

Verify that the signal $x(t) = \sin(t)$ is periodic.

$T = 2\pi/\Omega = 2\pi/1 = 2\pi$

$x(t) = x(t + kT),\ k \in \mathbb{Z}$

$1 \le k \le 10 \qquad 1 \le t \le 5.$

```
t=1:5;
x=sin(t)
T=2*pi;
```

```
x=0.84   0.91   0.14   -0.76   -0.96
```

```
for k=1:10
xk(k,:)=sin(t+k*T)
end
```

```
xk
```

```
xk =0.84   0.91   0.14   -0.76   -0.96
    0.84   0.91   0.14   -0.76   -0.96
    0.84   0.91   0.14   -0.76   -0.96
    0.84   0.91   0.14   -0.76   -0.96
    0.84   0.91   0.14   -0.76   -0.96
    0.84   0.91   0.14   -0.76   -0.96
    0.84   0.91   0.14   -0.76   -0.96
    0.84   0.91   0.14   -0.76   -0.96
    0.84   0.91   0.14   -0.76   -0.96
    0.84   0.91   0.14   -0.76   -0.96
```

The matrix xk is

$$xk = \begin{bmatrix} x(t+T) \\ x(t+2T) \\ x(t+3T) \\ \cdots \\ x(t+10T) \end{bmatrix}$$

2.4.1.1 Sum of Periodic Continuous-Time Signals

$x_1(t) = x_1(t + mT_1),\ m \in \mathbb{Z}$

$x_2(t) = x_2(t + kT_2),\ k \in \mathbb{Z}$

$x(t) = x_1(t) + x_2(t) = x_1(t + mT_1) + x_2(t + kT_2).$

$x(t) = x(t + T) = x_1(t + T) + x_2(t + T).$

$x_1(t + T) + x_2(t + T) = x_1(t + mT_1) + x_2(t + kT_2).$

If m, k are *prime* numbers $\Longrightarrow$ $mT_1 = kT_2 = T$ where $m, k \in \mathbb{Z}$.

2.4.1.1 Sum of Periodic Continuous-Time Signals

$x(t) = x(t+T) = x_1(t+T) + x_2(t+T)$.

If m, k are *prime* numbers $\Rightarrow$ $mT_1 = kT_2 = T$ where $m, k \in \mathbb{Z}$.

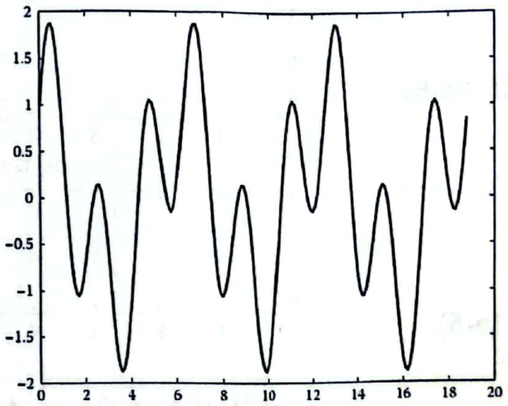
Example

Plot the signal $x(t) = \cos(t) + \sin(3t)$ in time of three periods.

$T_1 = 2\pi/\Omega = 2\pi$ $\Rightarrow$ $m = 1$ and $k = 3$ $\Rightarrow$ $T = 2\pi$

$T_2 = 2\pi/\Omega = 2\pi/3$

```
t=0:.1:6*pi;
x=cos(t)+sin(3*t);
plot(t,x)
```



Time definition ($T = 2\pi \Rightarrow 3T = 6\pi$).

The signal $x(t)$.

From the graph of $x(t)$ one can see that $x(t)$ is indeed periodic with period $T = 2\pi$.

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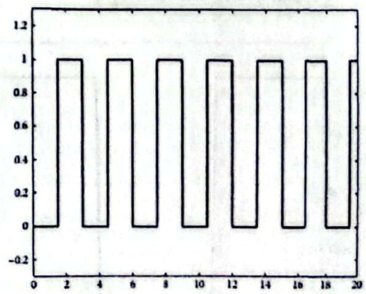
2.4.1.2 Construction of Periodic Signals

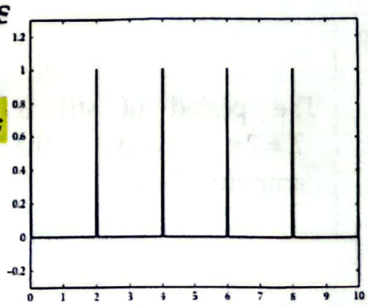
```
[s,t]=gensig(type,T,t,ts)
```

`type`	'sin'	sine signal
	'square'	rectangular periodic pulse
	'pulse'	periodic unit pulses
`T`	period	
`t`	time duration of the signal	
`ts`	spacing of the time samples	
	By default $ts = 10^{-3}$ s	

```
[s,t]=gensig('square',3,20,0.01)
plot(t,s)
ylim([-.3 1.3])
```



```
[s,t]=gensig('pulse',2,10);
plot(t,s)
ylim([-.3 1.3])
```



The signal is assigned to variable s, while time is assigned to variable t.

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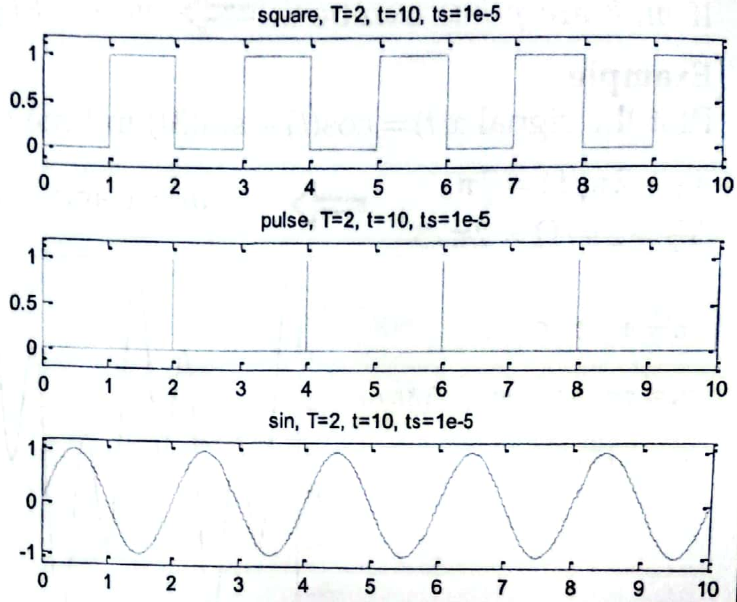
2.4.1.2 Construction of Periodic Signals

```
[s,t]=gensig(type,T,t,ts)
```

```
figure
subplot(3,1,1)
[s,t]=gensig('square',2,10,1e-5);
plot(t,s);ylim([-0.2,1.2])
title 'square, T=2, t=10, ts=1e-5'

subplot(3,1,2)
[s,t]=gensig('pulse',2,10,1e-5);
plot(t,s);ylim([-0.2,1.2])
title 'pulse, T=2, t=10, ts=1e-5'

subplot(3,1,3)
[s,t]=gensig('sin',2,10,1e-5);
plot(t,s);ylim([-1.2,1.2])
title 'sin, T=2, t=10, ts=1e-5'
```



متغیرهای t و s به صورت بردارهای ستونی ایجاد میشود.

2.4.1.2 Construction of Periodic Signals

x = square(t) $T = 2\pi/\Omega = 2\pi$.

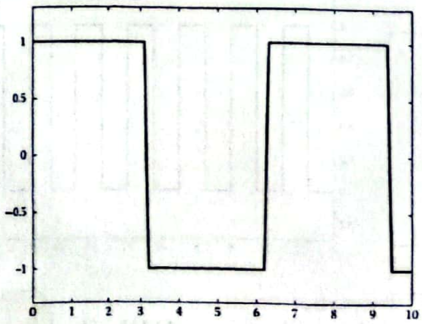
x = square(t,duty) duty, which is a number between 0 and 100.
The duty cycle is the percent of the period in which the signal is positive.

square(t) is similar to sin(t), but creates a square wave with peaks of ±1 instead of a sine wave.

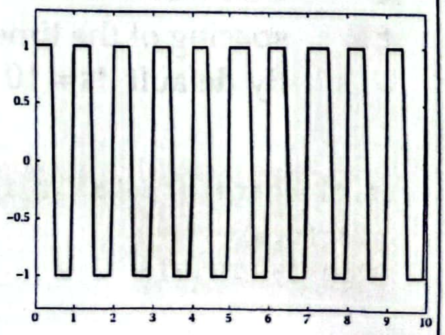
```
t=0:0.1:10;
s=square(t);
plot(t,s);
ylim([-1.3 1.3])
```

```
s2=square(2*pi*t);
```

The period of $s_2(t)$ is $T = 2\pi/\Omega = 2\pi/2\pi = 1$.



The period of $s(t)$ is $T = 2\pi$ and the amplitude is ±1.



2.4.1.2 Construction of Periodic Signals

sawtooth(t)
sawtooth(t,width)

sawtooth(t) is similar to sin(t), but creates a sawtooth wave with peaks of -1 and 1 instead of a sine wave. The sawtooth wave is defined to be -1 at multiples of 2π and to increase linearly with time with a slope of 1/π at all other times.

width, a scalar parameter between 0 and 1, determines the point between 0 and 2π at which the maximum occurs.

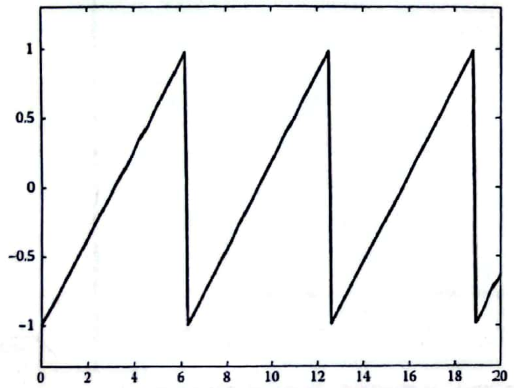
```
t=0:0.1:20;
s=sawtooth(t);
```

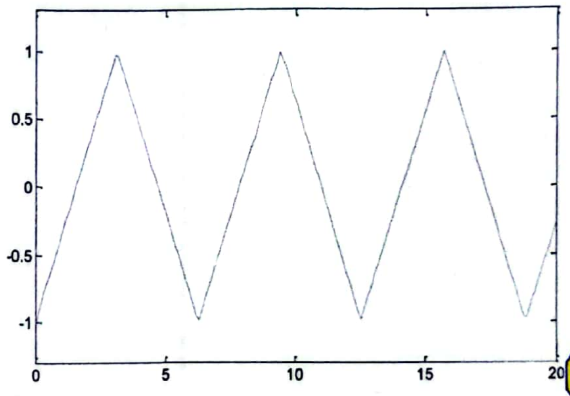
```
plot(t,s);
ylim([-1.3 1.3])
```

Definition of a triangle wave for $0 \le t \le 20$.

Indeed the period is $T = 2\pi$, while the signal takes values in $[-1,1]$.

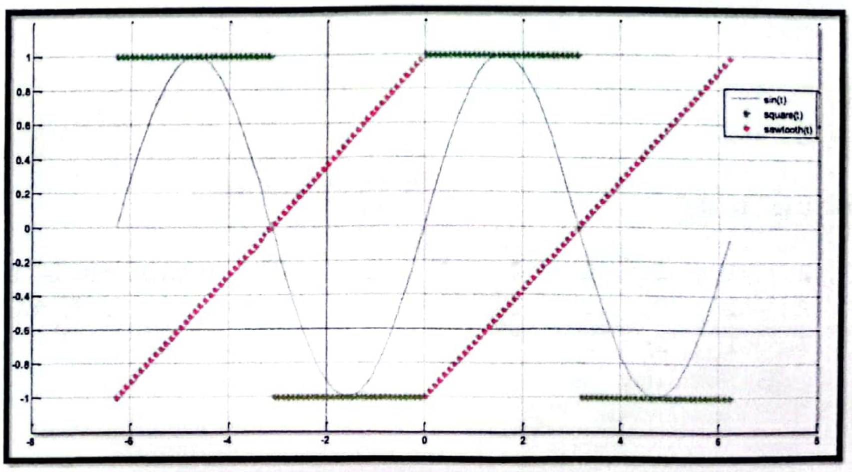
```
s=sawtooth(t,0.5);
plot(t,s)
ylim([-1.3 1.3])
```





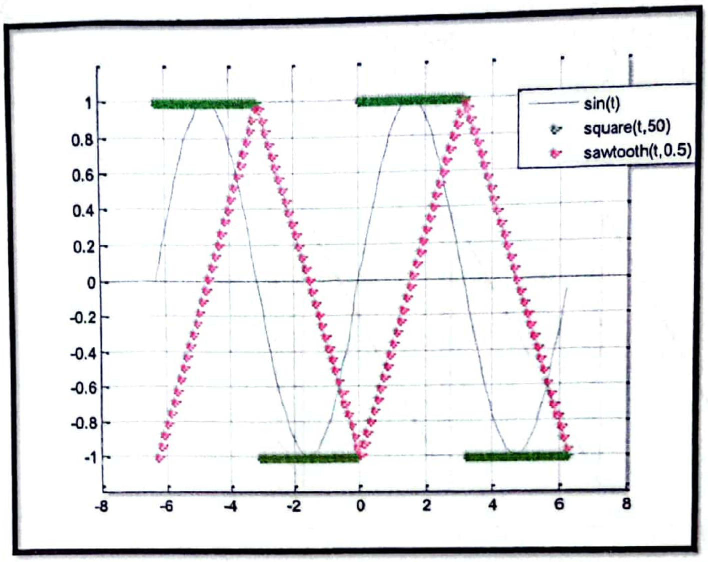
sin, square, sawtooth

```
clc; close all;
t=-2*pi:0.1:2*pi;
figure
hold on
plot(t,sin(t), t, square(t),'*', t, sawtooth(t),'*');
ylim([-1.2 1.2])
legend ('sin(t)', 'square(t)', 'sawtooth(t)')
grid
```



sin, square, sawtooth

```
clc; close all;
t=-2*pi:0.1:2*pi;
figure
hold on
plot(t,sin(t), t, square(t,50),'*', t, sawtooth(t,0.5),'*');
ylim([-1.2 1.2])
legend ('sin(t)', 'square(t,50)', 'sawtooth(t,0.5)')
grid
```



2.4.1.2 Construction of Periodic Signals

`B = repmat(A,sz1,sz2,...,szN)` specifies a list of scalars, `sz1,sz2,...,szN`, to describe an N-D tiling of `A`. The size of `B` is `[size(A,1)*sz1, size(A,2)*sz2,...,size(A,n)*szN]`. For example, `repmat([1 2; 3 4],2,3)` returns a `4`-by-`6` matrix.

```
x=[12];
N=3;K=4;
```

```
xp=repmat(x,N,K)
```

xp =	1	2	1	2	1	2	1	2
	1	2	1	2	1	2	1	2
	1	2	1	2	1	2	1	2

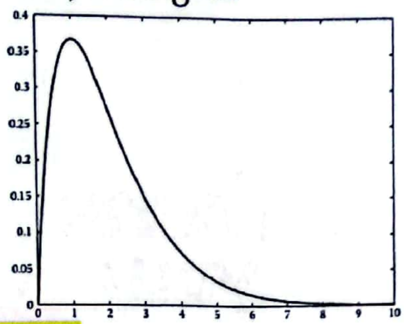
The matrix xp is a replica of vector x in N rows and K columns.

2.4.1.2 Construction of Periodic Signals

to repeat (suppose eight times) the signal

$x(t) = te^{-t}, 0 \le t \le 10.$

```
t=0:.1:10;
x=t.*exp(-t);
plot(t,x);
```



The signal $x(t)$ is defined over the time of one period (here $T = 10$).

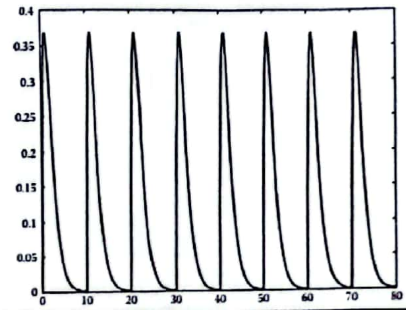
```
xp=repmat(x,1,8);
```

Vector x is replicated in 1 row and 8 columns.

```
tp=linspace(0,80,length(xp))
```

The time that the periodic signal xp will be plotted is estimated by dividing the time interval $0 \le tp \le 8T$ in a number of points equal to the number of elements of xp.

```
plot(tp,xp)
```



Graph of the periodic signal in the time interval $0 \le tp \le 80$. The signal $x(t) = te^{-t}$, $0 \le t \le 10$ is repeated for eight periods.

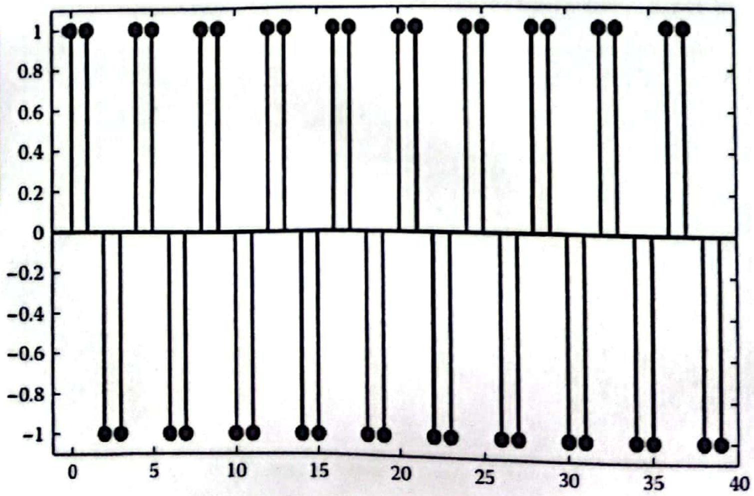
2.4.1.2 Construction of Periodic Signals

Example

Plot in 10 periods the periodic signal that in one period ($T = 4$) is given by

$$x[n] = \begin{cases} 1, & n = 0, 1 \\ -1, & n = 2, 3 \end{cases}$$

```
N=10;
x=[1 1 -1 -1];
xp=repmat(x,1,N);
n=0:length(xp)-1;
stem(n,xp);
```



The signal is repeated $N = 10$ times.

A discrete-time periodic signal can be also constructed easily by using the command `repmat`.

Thank you!